Composition Operators on Sobolev Spaces on Metric Measure Spaces

(joint work with S. K. Vodopyanov)

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Motivating questions

The simplest change-of-variables problem: I

Let $C[a, b]$ be a normed space of continuous functions on $[a, b]$. It is well known that the composition of continuous functions is continuous.

Consider the following converse question: Suppose that $\varphi\colon [a, b] \to [c, d]$ is a function such that $u \circ \varphi \in C[a, b]$ for every $u \in C[c, d]$. Is φ a continuous function?

The answer: φ is continuous. Indeed, take $u(x) = x$.

The simplest change-of-variables problem: II

Let AC^p, $1 \leqslant p < \infty$, be the seminormed space of absolutely continuous functions u such that $u' \in L^p$, $\|u\|_{AC^p} = \|u'\|_{L^p}$. Let $\varphi\colon [a,b] \to [c,d]$ be a homeomorphism.

We consider the change-of-variables problem on AC^p. Suppose that

$$\varphi^* u := u \circ \varphi \in AC^p[a,b] \quad \text{for every} \quad u \in AC^p[c,d],$$

and there exists a constant $K > 0$ such that the inequality

$$\|\varphi^* u\|_{AC^p[a,b]} \leqslant K \|u\|_{AC^p[c,d]} \quad \text{holds for every} \quad u \in AC^p.$$

We say that φ^* is a composition operator.

The simplest change-of-variables problem: III

Let AC^p, $1 \leqslant p < \infty$, be the seminormed space of absolutely continuous functions u such that $u' \in L^p$, $\|u\|_{AC^p} = \|u'\|_{L^p}$. Let $\varphi\colon [a, b] \to [c, d]$ be a homeomorphism. Suppose that φ^* is a bounded composition operator on AC^p.

Question

What geometric restrictions on φ follow from the boundedness of the change-of-variables operator? Are they sufficient?

Answer

For $p > 1$, φ must be Lipschitz. Determine $\|\varphi^*\|$.

For $p = 1$, every absolutely continuous φ induces a bounded composition operator. Determine $\|\varphi^*\|$.

Historical notes

Sobolev's conjecture

- In 1941, S. L. Sobolev proved that C^1 quasi-isometries preserve the $L^{1,p}$ spaces in $\mathbb{R}^n$, $n \geq 2$.
  He conjectured that all isomorphisms of Sobolev spaces $L^{1,p}$ are quasi-isometries.
- It follows from V. G. Maz'ya's PhD thesis that C^1 isomorphisms of $L^{1,p}$ spaces are induced by
  - quasi-isometries if $p \neq n$,
  - quasiconformal mappings if $p = n$.

Reshetnyak's problem

In 1968, Yu. G. Reshetnyak posed the problem of characterizing all isomorphisms of the Sobolev space $L^{1,n}$ induced by quasiconformal mappings.

- In 1974, E. A. Poletsky answered this question for the space $L^{1,n} \cap C^1$.
- In 1975, S. K. Vodopyanov and V. M. Gol'dshtejn gave a full solution of Reshetnyak's problem.

In the 1980s, the study of mappings that induce bounded composition operators on Sobolev spaces began.

Composition operators: I

Characterization of homeomorphisms that induce bounded composition operators on Sobolev spaces have been established in several model settings:

- on Carnot groups (including Euclidean space), see S. K. Vodopyanov and A. D. Ukhlov (1998), S. K. Vodopyanov and N. A. Evseev (2022);
- on Riemannian manifolds, see S. K. Vodopyanov (2024 and 2025).

The list of publications is not exhaustive; additional work includes results on measurable mappings and related topics.

Composition operators on metric measure spaces: I

Related results include:

- M. Mocanu (2006) proved that quasiconformal (QC) mappings on certain metric measure spaces (m.m.s.) induce continuous composition operators on Newtonian spaces $N_0^{1,Q}$;
- QC mappings induce bounded composition operators on Triebel–Lizorkin spaces, see P. Koskela, D. Yang, Y. Zhou (2011), and on Bessel potentials spaces, see M. Olivia, M. Prats (2017);
- Quasisymmetric mappings induce isomorphisms between Hajłasz–Triebel–Lizorkin spaces, see M. Bourdon, H. Pajot (2003);

Composition operators on metric measure spaces: II

Related results include:

- Partial necessary and sufficient conditions for the boundedness of composition operators on Hajłasz spaces were obtained by A. S. Romanov (2024, 2025);
- Homeomorphisms inducing bounded composition operators on Sobolev spaces on m.m.s. were characterized under additional assumptions; see A. Menovshikov, A. Ukhlov (2025).

The list of publications is not exhaustive; additional work includes results on composition operators on other functional spaces and related topics.

Preliminaries

Preliminaries: I

- A triple (X, d, μ) is a metric measure space if (X, d) is a nonempty separable metric space and μ is a nontrivial regular Borel measure.
  We also assume that μ is finite on bounded subsets.

Examples

- Euclidean spaces $(\mathbb{R}^n, d_{eu}, \mathcal{L}^n)$,
- Riemannian manifolds $(M^n, d_{riem}, \mathrm{vol}_g)$,
- stratified Lie groups (also known as Carnot groups) or sub-Riemannian manifolds,
- open domains in the above spaces,
- and many other.

Preliminaries: II

- The Dirichlet space $D^{1,p}(X)$, $1 \leqslant p < \infty$, consists of measurable functions $u\colon X \to \mathbb{R}$ such that the inequality
$$|u(x) - u(y)| \leqslant \int\limits_{\gamma} g\, ds$$
holds for all curves γ with endpoints $x, y \in X$. Here $g \in L^p(X)$ is a nonnegative Borel function independent of the curve γ and of the points x, y.
- Let $u \in D^{1,p}(X)$. There exists an L^p-minimal upper gradient. We denote it by $|\nabla u|_p$. The smallest upper gradient $|\nabla u|_p$ may depend on p.

For notational simplicity, we write $|\nabla u|$ instead of $|\nabla u|_p$.

For the theory of first-order calculus in m.m.s. see, e.g., J. Heinonen, P. Koskela, N. Shanmugalingam, J. T. Tyson (2015).

Preliminaries: III

- We say that a measurable mapping $\varphi\colon X \to Y$ belongs to the Reshetnyak class $D^{1,p}(X;Y)$, $1 \leqslant p < \infty$, if
  1. $u \circ \varphi \in D^{1,p}(X)$, for every $u \in \operatorname{Lip}(Y)$ with $\operatorname{Lip}(u;Y) \leqslant 1$, and
  2. there exists an L^p-envelope w such that

$$|\nabla(u \circ \varphi)| \leqslant w \quad \text{a.e.}$$

  The smallest envelope w is denoted by $|D\varphi|$.

$$X \xrightarrow{\varphi} Y$$

$$D^{1,p}(X) \xleftarrow{\varphi^*} \operatorname{Lip}(Y)$$

Main results

Characterization of composition operators

Theorem (S., 2025 ($p = q$); S. K. Vodopyanov, S., 2025, 2026 ($q < p$))

Let (X, d, μ) and (Y, ρ, ν) be metric measure spaces, and let $\varphi\colon X \to Y$ be a homeomorphism. Let $1 \leqslant q \leqslant p < \infty$.

The composition operator $\varphi^*\colon D^{1,p}(Y) \cap \operatorname{Lip}(Y) \to D^{1,q}(X)$ is bounded if and only if

1. *φ belongs to the Reshetnyak class $D^{1,q}_{\mathrm{loc}}(X; Y)$,*
2. *φ has finite distortion, i.e., $|D\varphi| = 0$ a.e. on the set $Z_\varphi := \{\mathcal{J}\varphi(x) = 0\}$,*
3. *the operator distortion function $K_{q,p}(\cdot, \varphi)$ belongs to $L^{\varkappa}(X)$, where $1/\varkappa = 1/q - 1/p$.*

Moreover, $\|\varphi^*\| = \|K_{q,p}(\cdot, \varphi)\|_{L^{\varkappa}(X)}$.

The proof of necessity for $q < p$: main ideas

Let $\varphi\colon X \to Y$ be a homeomorphism, and suppose that φ^* is bounded. Hence, for every $u \in D^{1,p}(Y) \cap \mathrm{Lip}(Y)$, $u \circ \varphi \in D^{1,q}(X)$ and

$$\|u \circ \varphi\|_{D^{1,q}(X)} \leqslant \|\varphi^*\| \|u\|_{D^{1,p}(Y)}.$$

Question

How can we extract geometric properties of φ?

A measure associated with the composition operator

We construct a finite regular Borel measure Ψ on Y such that the inequality

$$\int\limits_{\varphi^{-1}(B)} |\nabla(\varphi^* u)|^q \, d\mu \leqslant \Psi(B)^{\frac{q}{\varkappa}} \left(\int\limits_B |\nabla u|^p \, d\nu \right)^{\frac{q}{p}}$$

holds for every $B \in \mathscr{B}(Y)$ and every $u \in D^{1,p}(Y) \cap \operatorname{Lip}(Y)$. Here $1/\varkappa = 1/q - 1/p$.

It is clear that $\Psi(Y) = \|\varphi^*\|^{\varkappa}$.

The measure Ψ encodes **geometric properties** of the mapping φ.

Regularity of the homeomorphism: I

Fix a bounded open set $U \subset Y$. By the definition of Ψ, we have

$$\int\limits_{\varphi^{-1}(U)} |\nabla(\varphi^* u)|^q \, d\mu \leqslant \Psi(U)^{\frac{q}{\varkappa}} \Bigg(\int\limits_U |\nabla u|^p \, d\nu \Bigg)^{\frac{q}{p}}$$

$$\leqslant \Psi(U)^{\frac{q}{\varkappa}} \Bigg(\int\limits_U \operatorname{Lip}(u; U)^p \, d\nu \Bigg)^{\frac{q}{p}} = \Psi(U)^{\frac{q}{\varkappa}} \nu(U)^{\frac{q}{p}} \operatorname{Lip}(u; U)^q.$$

Regularity of the homeomorphism: II

$$\text{(previous slide)} \hookrightarrow \quad \int\limits_{\varphi^{-1}(U)} |\nabla(\varphi^* u)|^q \, d\mu \leqslant \Psi(U)^{\frac{q}{\varkappa}} \nu(U)^{\frac{q}{p}} \operatorname{Lip}(u; U)^q.$$

Hence, the operator

$$\varphi_U^* \colon \operatorname{Lip}(U) \to D^{1,q}(\varphi^{-1}(U))$$

is bounded.

We conclude that $\varphi \in D^{1,q}(\varphi^{-1}(U); U)$, and we have the inequality

$$\int\limits_{\varphi^{-1}(U)} |D\varphi|^q \, d\mu = \|\varphi^*\|^q \leqslant \Psi(U)^{\frac{q}{\varkappa}} \nu(U)^{\frac{q}{p}}.$$

In particular, since φ is a homeomorphism, it follows that $\varphi \in D^{1,q}_{\mathrm{loc}}(X; Y)$.

The homeomorphism has finite distortion

$$\text{(previous slide)} \hookrightarrow \quad \int\limits_{\varphi^{-1}(U)} |D\varphi|^q \, d\mu \leqslant \Psi(U)^{\frac{q}{\varkappa}} \nu(U)^{\frac{q}{p}}$$

Let $Z_\varphi = \{x \in X \mid \mathcal{J}\varphi(x) = 0\}$. Then $\nu(\varphi(Z_\varphi)) = 0$, and hence

$$\int\limits_{Z_\varphi} |D\varphi|^q \, d\mu \leqslant \Psi(U)^{\frac{q}{\varkappa}} \nu(U)^{\frac{q}{p}} = 0.$$

It follows that $|D\varphi| = 0$ a.e. on Z_φ.

From this, we can prove that Ψ is absolutely continuous with respect to ν. We denote the Radon–Nikodym density by Ψ'.

Integrability of the operator distortion function: I

(previous slide) $\hookrightarrow$ $$\int\limits_{\varphi^{-1}(U)} |D\varphi|^q \, d\mu \leqslant \Big(\int\limits_U \Psi' \, d\nu \Big)^{\frac{q}{\varkappa}} \nu(U)^{\frac{q}{p}}$$

For simplicity, suppose that φ has the Lusin N-property. Let $B \in \mathscr{B}(X)$ be a Borel subset, and put $U = \varphi(B)$. Using the change-of-variable formula, we obtain

$$\begin{aligned}\int\limits_B |D\varphi|^q \, d\mu \leqslant \Big(\int\limits_{\varphi(B)} \Psi' \, d\nu \Big)^{\frac{q}{\varkappa}} \Big(\int\limits_B \mathcal{J}\varphi(x) \, d\mu \Big)^{\frac{q}{p}} \\ = \Big(\int\limits_B \Psi'(\varphi(x)) \mathcal{J}\varphi(x) \, d\mu \Big)^{\frac{q}{\varkappa}} \Big(\int\limits_B \mathcal{J}\varphi(x) \, d\mu \Big)^{\frac{q}{p}}.\end{aligned}$$

A converse to Hölder's inequality: I

Let $1/p + 1/p' = 1$, $1 < p < \infty$.

Suppose that f, g, h are nonnegative measurable functions. If $f(x) \leqslant g(x)h(x)$ a.e., then

$$\int_B f\, d\mu \leqslant \left(\int_B g^p\, d\mu \right)^{\frac{1}{p}} \left(\int_B h^{p'}\, d\mu \right)^{\frac{1}{p'}}$$

for every measurable B (Hölder's inequality).

A converse to Hölder's inequality: II

Let $1/p + 1/p' = 1$, $1 < p < \infty$.

Theorem (A converse to Hölder's inequality)

Suppose that f, g, h are nonnegative measurable functions such that the inequality

$$\int\limits_B f\,d\mu \leqslant \left(\int\limits_B g^p\,d\mu\right)^{\frac{1}{p}} \left(\int\limits_B h^{p'}\,d\mu\right)^{\frac{1}{p'}}$$

holds for every measurable B. Then $f(x) \leqslant g(x)h(x)$ a.e.

Integrability of the operator distortion function: II

(previous slide) $\hookrightarrow$ $$\int\limits_B |D\varphi|^q \, d\mu \leqslant \Big(\int\limits_B \Psi'(\varphi(x)) \mathcal{J}\varphi(x) \, d\mu \Big)^{\frac{q}{\varkappa}} \Big(\int\limits_B \mathcal{J}\varphi(x) \, d\mu \Big)^{\frac{q}{p}}$$

By the converse to Hölder's inequality ($q/\varkappa + q/p = 1$) we obtain the inequality

$$|D\varphi|^q(x) \leqslant \big(\Psi'(\varphi(x))\mathcal{J}\varphi(x)\big)^{\frac{q}{\varkappa}} \mathcal{J}\varphi(x)^{\frac{q}{p}}$$

Hence, the operator distortion function has an $L^{\varkappa}$-majorant:

$$K_{q,p}(x,\varphi) := \frac{|D\varphi|(x)}{\mathcal{J}\varphi(x)^{\frac{1}{p}}} \leqslant \big(\Psi'(\varphi(x))\mathcal{J}\varphi(x)\big)^{\frac{1}{\varkappa}},$$

if $\mathcal{J}\varphi(x) \neq 0$. We set $K_{q,p}(x,\varphi) := 0$ if $\mathcal{J}\varphi(x) = 0$.

Integrability of the operator distortion function: III

$$\text{(previous slide)} \hookrightarrow \quad K_{q,p}(x,\varphi) \leqslant \big(\Psi'(\varphi(x))\mathcal{J}\varphi(x)\big)^{\frac{1}{\varkappa}}$$

By integrating $K_{q,p}(\cdot,\varphi)^{\varkappa}$ over X, we obtain the inequality

$$\int\limits_X K_{q,p}(x,\varphi)^{\varkappa}\,d\mu(x) \leqslant \int\limits_X \Psi'(\varphi(x))\mathcal{J}\varphi(x)\,d\mu(x) = \int\limits_Y \Psi'(y)\,d\nu(y) = \Psi(Y) = \|\varphi^*\|^{\varkappa}.$$

The inequality $\|\varphi^*\| \leqslant \|K_{q,p}(\cdot,\varphi)\|_{L^{\varkappa}(X)}$ follows from the proof of sufficiency.

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Q&A